

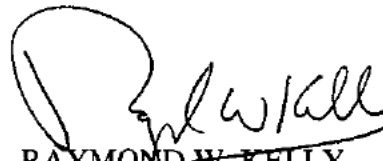


POLICE DEPARTMENT

-----X
In the Matter of the Disciplinary Proceedings :
- against - : FINAL
Police Officer William Rosario : ORDER
Tax Registry No. 903027 : OF
45 Precinct : DISMISSAL
-----X

Police Officer William Rosario, Tax Registry No. 903027, Shield No.18279 ,
Social Security No. ending in [REDACTED] having been served with written notice, has been
tried on written Charges and Specifications numbered 82283/06, as set forth on form P.D.
468-121, dated September 25, 2006, and after a review of the entire record, has been
found Guilty as Charged.

Now therefore, pursuant to the powers vested in me by Section 14-115 of the
Administrative Code of the City of New York, I hereby DISMISS Police Officer William
Rosario from the Police Service of the City of New York.


RAYMOND W. KELLY
POLICE COMMISSIONER

EFFECTIVE: 0001 Hours, 6/10/2008



POLICE DEPARTMENT

May 2, 2008

-----X
In the Matter of the Charges and Specifications : Case No. 82283/06

- against - :

Police Officer William Rosario :

Tax Registry No. 903027 :

45 Precinct :
-----X

At: Police Headquarters
One Police Plaza
New York, New York 10038

Before: Honorable David S. Weisel
Assistant Deputy Commissioner - Trials

APPEARANCE:

For the Department: Daniel Maurer, Esq.
Department Advocate's Office
One Police Plaza
New York, New York 10038

For the Respondent: Richard A. Dienst, Esq.
233 Broadway – 18th Fl.
New York, N.Y. 10279

To:

HONORABLE RAYMOND W. KELLY
POLICE COMMISSIONER
ONE POLICE PLAZA
NEW YORK, NEW YORK 10038

The above-named member on the Department appeared before me on March 12 and 13, 2008, charged with the following:

1. Said Police Officer William Rosario, assigned to the 45th Precinct, on or about and between June 5, 2006 through September 5, 2006, engaged in conduct prejudicial to the good order, efficiency or discipline of the Department in that said Officer did wrongfully ingest cocaine without police necessity or authority to do so.

PG 203-10 Page 1, Paragraph 5 – GENERAL REGULATIONS

2. Said Police Officer William Rosario, assigned to the 45th Precinct, on or about and between June 5, 2006 through September 5, 2006, engaged in conduct prejudicial to the good order, efficiency or discipline of the Department in that said Officer did wrongfully possess cocaine without police necessity or authority to do so.

PG 203-10 Page 1, Paragraph 5 – GENERAL REGULATIONS

September 5, 2006, [REDACTED]
New York City Police Department, without permission or authority to do so.

PG 203-10 Page 2, Paragraph 17 – GENERAL REGULATIONS

The Department was represented by Daniel Maurer, Esq., Department Advocate's Office, and the Respondent was represented by Richard A. Dienst, Esq.

The Respondent, through his counsel, entered a plea of Not Guilty to the subject charges. A stenographic transcript of the trial record has been prepared and is available for the Police Commissioner's review.

DECISION

The Respondent is found Guilty of Specification Numbers 1 and 2. The Department did not produce evidence concerning Specification Number 3, and upon its motion, made on March 13, 2008, Specification Number 3 was dismissed.

SUMMARY OF EVIDENCE PRESENTED

The Department's Case

The Department called Detective Shibhan Mockler, Sergeant Greacia Herdsman, and Doctor Thomas Cairns as witnesses.

Detective Shibhan Mockler

Mockler was assigned to the Drug Screening Unit (DSU) in the Medical Division as a hair sample taker. She had performed over one hundred sampling procedures in her year at DSU, which began about February 2006. Mockler was "certified" after receiving training in the form of a one-hour lecture given by "the drug company," reading literature, and passing an examination. She had also received a similar lecture during a previous stint at the Medical Division when she took hair samples from police candidates.

Mockler testified that the Respondent was brought in for a random drug test, or Dole test, on September 5, 2006, although she did not specifically recall taking a sample of the Respondent's hair. She described the general Dole procedure. The member of the Department being tested would enter his name, tax number and command into a log, along with any substances ingested in the previous three months. The member would sign the form after presenting Mockler with his Department identification. The member

would also place a fingerprint on the form (see Department's Exhibit [DX] 1, Drug Screening Questionnaire for Resp't, dated Sept. 5, 2006, donor no. 6-2827-06-XH).

Mockler testified that a donor number would be placed on the chain of custody form; this would also be signed and dated by the member. Mockler would place the donor number on the sample pouch, and would initial and date this.

Mockler and the member being tested would enter the cutting room. Mockler testified that the member was "isolated" in the cutting room, and said, "No one is there but me." She admitted, however, that occasionally more than one tester-donor pair would be in the cutting room at any one time. The cutting room was about 15 feet by 8 feet, and approximated the corner of the Trial Room where the witness stand was located. There were two cutting desks on opposite walls of the room. Each desk was about the size of the witness stand. Only the person cutting the hair would stand next to the member being tested.

Mockler testified that she would spray the table with alcohol and lay white medical paper on it. She would place the paperwork on the table, along with the sample pouch, the foil, and the security envelope. Mockler would wipe the scissors with alcohol, although she testified that a new pair of scissors was used for each person. Mockler did not wear rubber gloves, but did rub sterilizing foam on her hands "before and after each cut."

Mockler stated that she would cut three samples of hair from the member's head. She would cut the samples from the same area of the head. Whenever possible, Mockler would cut from the "crown," which she described as the top of the rear of the head.

Each sample would be wrapped in foil and placed in an individual pouch. Red security seals, dated and initialed by the member, would be placed on each pouch. Each pouch would be put into a security envelope, also signed and dated by the member.

Mockler testified that the first and second samples were each given a chain of custody form (see DX-2 & DX-3, chain of custody forms), and that the third sample would be secured at the Sick Desk. Each chain of custody form contained a certification by the member that the hair placed with each form was his hair.

Mockler stated that the Respondent's samples were sent out for testing by Psychemedics, and the results came back positive for cocaine.

Mockler testified that she took about ten random Dole samples per day.

Sergeant Greacia Herdsman

Herdsman was the Commanding Officer of the Dole unit. She testified that when her unit received a report of a positive test, the third sample maintained by the Medical Division was placed in a safe at the Sick Desk. Herdsman stated that if a member wanted the third sample to be tested, he was supposed to make a written request to the Department Advocate's Office, specifying a laboratory of his choosing. The member would then call Herdsman to inform her that he would be picking up the sample.

Herdsman testified that the Respondent made a request through his attorney at the time that the third sample be tested by Quest Diagnostics (see DX-4, letter dated Oct. 11, 2006). The third sample was sent to Quest on October 18, 2006 (see DX-5, Quest chain of custody sheet, donor no. 6-2827-06-XH).

Herdsmen described the procedure used to send out the third sample. The sample would be removed from the envelope in the tested member's presence, and Herdsmen would "allow him to look at the initials on the forms, to make sure that that is his." The transport folder would be sealed in his presence. The member had the option of waiting for the courier to pick up the sample.

Herdsmen testified that she received the results of the test on the Respondent's third sample (see DX-6, Quest Diagnostics test result form). The test was conducted on October 24, 2006, and indicated that the results were positive for cocaine and its metabolite at levels above the Department's cutoff. Herdsmen's understanding was that anything below the cutoff level could not be used for prosecution.

Herdsmen testified that all third samples for a given day were kept in a single manila envelope, which would be signed and sealed by the securing officer. She averred that the Respondent's was the only third sample for September 5, 2006, that tested positive.

Doctor Thomas Cairns

Cairns was stipulated to be an expert in forensic toxicology. He held a Ph.D. in analytical chemistry from the University of Glasgow, and the higher degree of Doctor of Science in toxicology, awarded by Glasgow's Faculty of Science. Cairns had about 100 to 150 publications and conference presentations to his credit. Some of his presentations specifically were about drug testing of hair. He had testified on several occasions as an expert, and dozens of times regarding mass spectrometry.

Cairns testified that in 1987, while the director of the federal government's National Center for Toxicological Research, he was approached by the Psychomedics Corporation of Culver City, California, to develop methods for drug testing through hair. He became a full-time employee of Psychomedics in 1995, and was currently their senior science advisor. His role was to review the documentation to be used in courts in order "to ensure myself that the chain of custody was intact, that all other analytical controls and checks and balances were adhered to." Cairns asserted that Psychomedics had performed over five million hair tests, beginning in 1988. He estimated that he had overseen hundreds of thousands of hair tests.

Cairns testified that the Psychomedics laboratory in Culver City was certified by New York State. They were also licensed by the federal Department of Health and Human Services, and approved by the Food and Drug Administration. Cairns testified that it was the FDA's role to "review and evaluate the technology used to provide a clearance which states that the methods used are precise, and accurate and reliable for the purpose of drug detection in hair." The machines used by Psychomedics are tested and calibrated. Cairns admitted that the Department was a major client of Psychomedics, and that he has always appeared as a Department witness when in the Trial Room.

Cairns asserted that when a person ingests cocaine, it enters the bloodstream, passes through the vital organs, and becomes metabolized. A metabolite, Cairns testified, was a molecule derived from a principal compound, like cocaine, but whose structure is changed to become water soluble, and it thus may be excreted from the body through urine. The principal metabolite of cocaine was benzoylecgonine (BE). BE was a metabolite specific to cocaine.

Aside from being eliminated in urine, Cairns testified, BE also enters the hair follicle. Cairns testified that the follicle grows the structure of the hair, which is made of keratin, by using compounds from the bloodstream. He described keratin as a “polymeric” material that, as it builds from the bloodstream, also traps cocaine and BE molecules that are present in the bloodstream. The sebaceous gland generates oil, which lubricates the skin and allows the hair to break through the skin.

Cairns stated that “the dermatological societies” generally accept that hair grows about half an inch per month, so that every half-inch of hair becomes “an approximate thirty-day window” that displays “what drugs are circulating in the blood as the hair grows.” Cairns testified that the general scientific consensus was that cocaine could be detected in the urine up to approximately 72 hours in the past.

Cairns described the general protocol at Psychemedics. First, the seal on the sample pouch is examined for tampering. The pouch is then opened, and the sample acquisition container, which also has a seal, signed by the donor and collector. If “the paperwork is correct,” the sample is given a unique barcode, and then is tested.

Cairns stated that if the hair strands are longer than 3.9 cm, they are cut to that length. There are generally about eighty 3.9-cm-length strands in each sample, weighing a total of approximately 40 milligrams. Eight milligrams are used for the test, with the remainder set aside for confirmation if that becomes necessary.

The hair is then liquefied, releasing any drugs from the keratin, and separated into five test tubes for the five drug types being scrutinized: cocaine, opiates (such as codeine), PCP, amphetamines, and marijuana.

Cairns stated that his lab first checks for a “presumptive positive” using the “screening method” of radioimmunoassay (RIA). A presumptive positive means that RIA detected a concentration of cocaine at “about equal to or above the cutoff.”

Cairns testified that the “administrative cutoff” was a level of compound, set conservatively “by clinical analysis that clearly differentiates a user from a nonuser, or one who may have been passively exposed to the drug.” He asserted that passive ingestion, such as at a rock concert, would be “far below the administrative cutoff.”

Cairns stated that there were two parts to the administrative cutoff used by the Department. First, the concentration of cocaine must be at or above 5 milligrams, per 10 milligrams of hair. There are 1,000 picograms (pg) in 1 nanogram (ng), 1,000 nanograms (ng) in 1 microgram, and 1,000 micrograms in 1 milligram. Second, the concentration of the BE metabolite must be 5 percent or greater of the concentration of cocaine.

Cairns testified that RIA is performed by making an antibody of cocaine radioactive and observing whether it “lock[s]” with any cocaine present in the sample. The radioactive compound can then be measured to determine the concentration of cocaine.

Cairns testified that if RIA revealed a presumptive positive, 12 additional milligrams of hair are taken from the sample in order to perform mass spectrometry (MS). If MS determined that the concentration was below the administrative cutoff, the sample would be recorded as negative.

Cairns defined mass spectrometry as “fingerprinting” substances at a molecular level, “the theory being held that each molecule in the universe has a unique fingerprint.” MS gives each molecule “an unambiguous qualification.” The molecule is divided into

its ions, which are then re-composed to "form the picture" of the molecule. A detector in the MS machine ascertains how many molecules are present at the correct molecular weight of cocaine. The smallest amount of compound by which MS could report a concentration was called the limit of detection.

Cairns asserted that the 12 milligrams of hair is "washed aggressively" with isopropanol to remove cosmetics. Then, the hair is subjected to five washes with a phosphate buffer; these washes in toto last three and a half hours. The hair is liquefied and sent for MS.

If the MS comes back above the administrative cutoff, Cairns stated, the fifth wash is examined by RIA to see whether the hair was truly decontaminated. Cairns stated that "in many cases," the final wash reveals zero cocaine, meaning that any cocaine or BE deposited through sweat or "environmental exposure" was removed.

Cairns stated that the records in a "laboratory data package" were "a complete chain of custody, meaning anyone who touched the sample and performed any analytical procedure, will indicate what they did with the sample, and initial the document." He prepared the packet for subject number 6-2827-06-XH, and certified the results (see DX-7, Psychomedics Laboratory Data Package). Cairns testified that the results of the Respondent's hair tests were reported under this subject number.

Cairns stated that the first sample of the Respondent's hair was positive for cocaine at a concentration of 6.5 ng of cocaine per 10 mg of hair. The sample was also positive for BE at 0.9 ng per 10 mg of hair, which was more than 5 percent of the 6.5 ng cocaine concentration. The fifth wash contained zero cocaine or BE. The first sample

was 3.6 centimeters long. Because 1 ½ inches equaled 3.9 cm, and every half-inch of hair represented about 30 days, the sample looked back on about 80 days of growth.

The second sample of the Respondent's hair, 2.7 cm long, was positive for cocaine at a concentration of 5.3 ng of cocaine per 10 mg of hair. The sample was also positive for BE at 0.6 ng per 10 mg of hair, which was more than 5 percent of the 5.3 ng cocaine concentration. The last wash again contained zero cocaine or BE. This second sample was 2.7 cm long. One inch equaled 2.6 cm, so the sample represented about 60 days.

Based on the results, Cairns concluded that the Respondent ingested cocaine recreationally "on at least more than half a dozen times." These results would not appear, Cairns opined, if the subject did only one line of cocaine.

Cairns asserted that the significance of the fifth wash being zero was that any cocaine present came from what was trapped inside the hair follicle after ingestion, not external contamination.

Cairns testified that the third sample of the Respondent's hair was tested by Quest Diagnostics (see DX-6, Quest laboratory report). The Quest results were 520 pg of cocaine per milligram of hair, or 5.2 ng of cocaine per 10 mg of hair. Quest also found 101 pg of BE per milligram of hair, or 1.0 ng of BE per 10 mg of hair. These results, Cairns opined, also demonstrated multiple ingestions of cocaine.

Cairns testified that the Respondent challenged whether the hair tested by Psychemedics was actually his hair. Psychemedics was contacted by Mitotyping Technology, and sent a hair sample to them for the purposes of performing mitochondrial DNA testing.

Cairns acknowledged that the concentration of cocaine in a “regular, active” user “could be quite high.” He stated that if the “millions of samples that I’ve done” that tested positive were looked at as a bell curve, the mean would be between 10 and 20 nanograms, and at “the far end,” a “couple of hundred.” He admitted that the difference between the negative value of 4.9 ng and the Respondent’s 5.3 ng reading was “infinitesimal” when viewed as a percentage.

Cairns denied that the term “false positive” meant “that perhaps people who have .50 above the cutoff mark have been accused of utilizing the cocaine, when, in fact, . . . they do not use cocaine.” Instead, he defined false positive as when the radioimmunoassay screening states that the sample is presumptive positive, but mass spectrometry fails to confirm this. Such a result would be reported to the Department as negative. Cairns denied that there had been instances where passive absorption of cocaine had resulted in false positives.

Cairns denied that passive contact with cocaine through transdermal absorption, similar to what happens with a nicotine patch, could cause transmission of cocaine into the polymeric keratin structure of the hair.

Cairns denied that if someone was in an area with extensive cocaine use, and scratched his head, and “it went into the subcutaneous layer, and was in . . . the hair itself for a period of time,” this would be “a matter of absorption, other than voluntarily ingested through either smoking it, or injecting it, or snorting it.” Chemicals from the outside environment could penetrate the epidermis, “[b]ut the washing procedure is meant to remove all cocaine that could have made its way there from the environment.” Cocaine that was “scratched in could not be trapped into the polymeric keratin structure.”

Cairns contended that “drugs ingested, making its way to the bloodstream, feeding the keratin polymeric material is trapped.” External contaminants that “enter the mature hair follicle after it leaves the surface of the head may be incorporated, or lay beneath the epidermis.” This external taint is “not trapped, and can be easily removed by washing.”

Cairns asserted that passive cocaine ingestion could not cause a positive test result, and he believed “the general consensus is that passive is taken care of by the application of a cutoff.”

Cairns stated that getting a haircut was an “avoidance tactic” for hair testing. Cairns claimed that the Respondent had gotten a haircut before going for an independent hair test because he only had 0.25 to 0.50 inches of hair on his head when he went for that test. He asserted that forensically, if one is challenging a positive test result by going to another lab, the individual should state this to the lab so that the lab will test at the limit of detection, rather than use an administrative cutoff.

Cairns also stated that while a narcotics officer “could be” at a greater risk of testing positive, a peer-reviewed study was published by an epidemiologist named Michewosky, in which eighty undercover narcotics officers from Pinellas County, Florida, were tested, and none were positive above the administrative cutoff. Cairns denied that a person’s skin color or ethnic background could affect the cocaine or BE test results.

The Respondent’s Case

The Respondent called Doctor Ronald B. Hoffman as a witness, and testified on his own behalf.

Doctor Ronald B. Hoffman

Hoffman was an adjunct associate professor at the Forensic Science Department of John Jay College of Criminal Justice, and an adjunct professor at Pace University's Forensic Science Program. Hoffman also had a private consultancy practice in forensic toxicology. He was deemed an expert in forensic toxicology.

Hoffman held Master's and Doctoral degrees in chemistry from Columbia University, and did a post-doctoral training fellowship in pharmacology at New York University Medical Center. He was board-certified by the American Board of Forensic Toxicology. He spent 27 years as a toxicologist at the Office of the Chief Medical Examiner of New York City. He had lectured on several occasions to police and fire investigators, and various legal and medical groups, including medical students and at hospitals. Hoffman had co-authored several publications on toxicology, but none "specifically" regarding hair testing. He had testified in the past as an expert in the Trial Room approximately two times, and on other occasions at the Office of Administrative Trials and Hearings. Hoffman admitted that he had never performed workplace hair testing, but was "familiar with the basic principles that are involved," such as "how drugs get into hair, and . . . the techniques that are involved in analysis."

Hoffman averred that the analytic aspects of hair testing "would be considered accurate" if the mass spectrometer were functioning properly. Hoffman indicated that the mass spectrometer actually gives a series of possible compounds that the tested substance might be, along with different levels of probability.

Any test, however, is subject to "experimental variation," Hoffman testified. There is human error, and normal variation in the instruments, "meaning that if you do

the exact same thing two times or three times, you are not going to get exactly, precisely the same number.” Hoffman asserted that strands of hair taken from one part of the scalp may have “a slightly different content of drug” than other parts, and that there was evidence that dark hair incorporates more drug than light hair.

Hoffman stated that there were two types of cutoffs with regard to hair testing. The analytic cutoff was “the limitations in terms of detectability of the instrument,” while the administrative cutoff was somewhat higher. A “regular cocaine user” would have a result “very, very significantly higher” than the administrative cutoff, with triple-digit results in “many cases.” Hoffman contended that a reported level near the administrative cutoff “would be considered, by definition, a trace amount,” especially compared to a “heavy” user.

Hoffman claimed that a result of 5.3 nanograms could be the result of experimental variation, and that there was no toxicological difference between that and the cutoff. 6.5 ng was also very close to the cutoff.

Hoffman testified that drugs could enter hair in several ways. One was to enter the hair follicle and then the strand as the hair grows. The drugs flow from the capillaries that nourish the follicle into the strand. It is also “known that drugs can be incorporated into hair as a result of hair coming into contact with the secretions of sweat glands and oil glands that would be present in the scalp,” although this “may not be the major route.” If there is “drug dust, or some type of drug material in the atmosphere,” and it settles onto the hair, it may dissolve in the sweat or oil secretions. It can then be “incorporated, to some degree, into the hair,” because at least part of the hair contacts the secretions.

Drugs can also be absorbed transdermally, in which drugs enter into the bloodstream through the skin. Hoffman testified that if someone scratched his scalp, "if there is a sufficient breach in the integrity of the skin, that would move in the direction of . . . facilitating transport of drug via the dermal route."

Hoffman stated that most United States currency, between 75 and 90 percent, is contaminated with very small amounts of drugs. The amount is generally in the microgram range. Hoffman stated that someone that handled just a few bills in a day would not be expected to return a positive result. This individual would have to be "a very, very heavy handler of currency," as opposed to "one or two bills a day." The person would have to "during the day routinely" be "counting out money," re-sorting, or distributing it. It would be someone that handled hundreds or perhaps thousands of bills each day, beyond a "typical cashier's contact."

Hoffman stated that he had performed work on the area of currency contamination, but admitted that no studies had been done on hair testing of individuals that handled large amounts of currency.

Hoffman contended that it could not be stated with a reasonable degree of scientific certainty that an individual who "handles very large amounts of currency, . . . much, much greater than . . . the average person," and considering that once in a while, there is a bill with a larger than normal amount of drugs present, that a positive result could not have come from handling currency.

Hoffman agreed that external contamination would not lead to a positive test result if one considered only the transfer of drugs from the capillaries to the follicle, and from the follicle to the hair shaft. He contended, however, that drugs could get into

surface sweat or oil, or if someone had drugs on his hand then touched his scalp. This “alternative route” does not feed drugs into the follicle. It, however, “does incorporate drug” into the hair strand, and washing would not remove it. For example, analyses had been performed in which drugs entered hair after the hair was soaked in a drug solution.

Hoffman asserted that “you would expect that drugs that are able to make their way from externally to the internal part of the hair shaft, . . . once it’s inside, that it would bind similarly to the way drugs that get in there from the follicle route.” The method in which the drug is absorbed “is still under investigation,” Hoffman admitted, and “not entirely worked out.” Some believed that melanin plays a role.

Hoffman added that there was “controversy and discussion” among toxicologists as to whether all wash procedures are equally effective in removing drugs. He asserted that if the first wash was negative, one could state that there was no external contamination “at least in the immediate past.”

Hoffman agreed that it would not be expected that an individual would test at or above the cutoff if he was at a nightclub where others were using cocaine, and there was cocaine smoke or powder in the air. He averred that “in most circumstances, casual contact would certainly not be expected to exceed the cutoff.” The same would be true if someone touched cocaine with his hands, and then scratched his head; Hoffman would not expect to see a positive result, whether through the skin in the hand or into the hair on the scalp.

Hoffman testified that Medtox Labs conducted a test on hair of the Respondent collected on September 22, 2006, and the results were negative for all drugs (see Respondent’s Exhibit [RX] B, Medtox report). A test performed by BioReference

Laboratories on the Respondent's urine on September 21, 2006, also came back negative for all drugs (see RX-A, BioReference report).

An immunoassay screening test was also performed by Omega Laboratories (see DX-10, Omega report). The hair was collected on September 22, 2006, at a length of less than 0.25 to 0.50 inches, meaning that the hair represented between zero and thirty days. The results were negative "for the five-drug pattern."

Hoffman testified that general laboratory procedure was that unless "some special arrangements" are made, mass spectrometry is not performed once the screening is negative.

The Respondent

The Respondent was 40 years old. He indicated that he was separated from his [REDACTED] but provided support for his [REDACTED]. He was appointed to the Department on January 19, 1993. He had one Excellent Police Duty medal, and was named Cop of the Month in August and September of 2005. He had worked as the summons officer, operating in a scooter.

The Respondent testified that he never used cocaine while a member of the Department, and did not know how it was present in his hair. He stated that he did not drink or smoke. When he was called for a random drug test at the Medical Division for September 5, 2006, he did not seek to postpone it. The Respondent agreed that the Department's witnesses from the Medical Division testified accurately as to the procedures used during his test.

The Respondent stated that when he was informed on September 22, 2006, of his positive result, he was "stunned." He told the Internal Affairs Bureau that he never took drugs in his life, and offered to undergo a lie detector test. He was still willing to take one at the time of trial.

After he received his Department hair test results, the Respondent said, he called his [REDACTED] who worked for a law firm. She advised him to get a second opinion, and gave him the names of several laboratories.

The Respondent went to one lab in Manhattan on the same day that he received his positive Medical Division result, and one in The Bronx the next day. He paid for these tests himself, between \$40 and \$100 each, because he was told he "had no coverage anymore" after he was suspended. He went to these labs because he had never done drugs in his life and wanted to prove his innocence.

The Respondent said that Omega Laboratories took his hair samples from the same place in the back of the head that the Medical Division used. He told Omega that he wanted a drug test, but he did not tell them that he had already failed a test and was seeking confirmation. He did not tell Medtox Labs either that he wanted a confirmatory test. The Respondent denied that when he went to Omega, there were missing patches of hair on his head.

The Respondent denied that he got a haircut in the seventeen days between his Medical Division sampling and his Omega sampling.

The Respondent testified that he also took a blood and urine test after receiving the positive results from the Medical Division. He took blood, hair, and urine tests because he "didn't know what he needed to take, I just took all three." He asserted that

the date on the report of September 21, 2006 (see RX-A, BioReference Laboratories report), was erroneous.

The Respondent admitted that his third Medical Division sample, which he asked be tested by Quest Diagnostics, came back positive. He also admitted that a DNA analysis by Mitotyping Technology revealed that the hair maintained by Psychemedics from the first and second samples was his hair.

The Respondent testified that his [REDACTED] owned a soda and snack vending machine company with about thirty machines in The Bronx, Manhattan, and Brooklyn. He did not know the name of the company, and did not even know if it had a name. He stated that the company was run out of [REDACTED] in Manhattan. He described the neighborhoods where the machines were located as "[l]ower class, like in the Bronx, in Brooklyn," and Washington Heights, "not the greatest areas."

The Respondent said that "pretty much the month of August and part of September" 2006, he helped his [REDACTED] count the money five or six times a week. "She can't handle it, she also has another job working for a cab company," and the machines were busier during the summer. He helped his [REDACTED] on other occasions besides August and September 2006. The Respondent would count between \$500 and \$800 per day, all in one-dollar bills. He was not sure if the machines get counted every day.

On cross-examination, the Respondent admitted that he said in his Official Department Interview that he had gone on a two-week vacation in August 2006 to Cancún. He insisted that on direct examination, he "said part of August, part of September."

After separating from his [REDACTED] the Respondent testified, he went to dance clubs, “hanging out with the guys.” To his knowledge, no one was using cocaine there.

The Respondent testified that prior to Hoffman’s testimony he “had no idea about” how cocaine-tainted money could get onto one’s head, then the hair. He did some Internet research after his positive result on what could cause a test to come back positive, but not how to beat the tests. He asserted that he did not find out anything.

FINDINGS AND ANALYSIS

Specification Nos. 1 & 2

The Respondent is charged with possessing and ingesting cocaine without police necessity or authority after a random Dole test revealed the presence of cocaine in his hair. The Court finds the Respondent Guilty of both specifications.

The Court credits the expert scientific testimony provided by Dr. Thomas Cairns of the Psychomedics Corporation. He explained that when cocaine is deliberately ingested, it metabolizes in the bloodstream, and is then fed to the hair, where it becomes trapped in the polymeric keratin hair structure. Also trapped, as a result of the body’s metabolizing the cocaine, is benzoylecgonine (BE). Cairns testified that BE can only be present as a result of the metabolism of cocaine by the body’s organs. Cairns also stated that both cocaine and BE were present in the Respondent’s hair at levels above the administrative cutoff. In fact, Cairns opined that the results meant that the Respondent had recreationally used cocaine more than six times, at least.

While the defense expert, Dr. Ronald B. Hoffman, was credible in some respects, his testimony as a whole was insufficient to rebut the Psychomedics evidence. First,

Hoffman testified that human and mechanical error and variation can occur within the laboratory process. Nevertheless, Cairns reviewed the Psychomedics laboratory data package prepared for the Respondent's tests, and certified that, in fact, their standard operating procedures were properly followed.¹

Hoffman also testified that the result on the second sample, 5.3 ng of cocaine per 10 mg of hair, was too close to the 5.0 ng cutoff to be reliable. This assessment begs the question of how far above the administrative cutoff a result must be to lead to a guilty finding. Cairns testified that the cutoff is set to ensure that a positive test result is recorded only for those that deliberately ingested cocaine. It "is a concentration level established by clinical analysis that clearly differentiates a user from a nonuser, or one who may have been passively exposed to the drug," such as by secondhand smoke. The cutoff is picked "with a wide margin of conservatism." In other words, someone who tests at 5.0 ng is not just *questionably* a recreational, deliberate drug user. Rather, that conclusion can be conservatively stated. It follows that the Respondent's 5.3 ng result, while only 0.3 nanograms greater than the cutoff, still conclusively establishes that he deliberately ingested cocaine. Otherwise, no matter how high the administrative cutoff is set, one could claim that his result was too close and could not be considered reliable proof. Accordingly, the Court rejects the Respondent's argument and finds that the Department proved through the Psychomedics and Quest test results that the Respondent ingested cocaine. See Matter of McBride v. Kelly, 215 A.D.2d 161 (1st Dept. 1995) (substantial evidence that officer ingested cocaine was provided by immunoassay and mass spectrometry).

¹ It was not necessary for the Department to produce the individual Psychomedics laboratory technicians that performed the testing on the Respondent's samples. See Matter of Gordon v. Brown, 84 N.Y.2d 574, 576, 578-80 (1994).

One key area of disagreement between Cairns and Hoffman was whether cocaine that lands on the hair or scalp can penetrate the hair shaft and be trapped within the hair structure. Cairns asserted that the only way for cocaine and BE to be present in the shaft is for those compounds to be trapped within the keratin structure through ingestion. As the drug makes its way through the bloodstream, Cairns testified, it is metabolized, and both the cocaine and BE enter the hair, becoming trapped in the keratin as the shaft grows. Hoffman, however, asserted that cocaine could fall onto the hair or scalp, dissolve in oil or sweat secreted by the sebaceous glands, and then could enter the hair shaft and become trapped within the keratin.

On this point, the Court credits the testimony of Cairns over that of Hoffman. While both are well-credentialed forensic toxicologists, Cairns has more experience with hair testing for abused drugs, as opposed to the practice of forensic toxicology in general. Notably, Hoffman admitted that studies had not concluded *how* cocaine could enter and be trapped in hair in this external contamination scenario. It is more reasonable that cocaine is trapped in keratin as Cairns described it: as the keratin structure grows from the bloodstream, the cocaine and BE come from the bloodstream also and are trapped with the keratin.

Nevertheless, the testimony of Hoffman and the Respondent, taken together, was offered to suggest that cocaine might have entered the Respondent's hair when he was helping his sister count money for her vending machine company. The proffered scenario was that the Respondent touched cocaine present on the currency, and then touched his scalp, providing a conduit into the hair.

Assuming arguendo the plausibility of this scenario, however, the fact remains that not only cocaine was present in the Respondent's hair, but also the BE metabolite. The presence of BE means that cocaine was metabolized, i.e., processed by the body's vital organs and changed into other compounds. This means that the cocaine was ingested by the Respondent and did not simply land on and enter his hair.

Indeed, the only other scenario Hoffman posited in which cocaine could have entered the bloodstream unbeknownst to the individual being tested was one inquired about by the Court. The Court asked Hoffman what could happen if "someone scratches their scalp." Hoffman replied that there would have to be "a sufficient breach in the integrity of the skin" for this to "move in the direction of causing . . . facilitating transport of drug via the dermal route." Hoffman thus suggested that the skin would have to be broken in order for the drug to enter the bloodstream. The Respondent did not testify, however, that anything close to that occurred.

Accordingly, the presence of BE in the Respondent's hair proves that the Respondent ingested cocaine, and that the cocaine passed through his vital organs, producing BE and a remainder of cocaine. These two compounds were then fed from the bloodstream into the Respondent's hair follicles, and then into his growing hair. The Department's proof negates the possibility that the positive test results occurred through external contamination of the Respondent's hair. See Matter of Connor v. New York City Police Dep't, 22 A.D.3d 425 (1st Dept. 2005) (there was substantial support for trial commissioner's finding that amount of cocaine metabolite in the officer's urine was inconsistent with claim of unknowing ingestion); Matter of Green v. Sielaff, 198 A.D.2d 113 (1st Dept. 1993) (where correction officer and bartender testified that officer was at a

bachelor party where liquor was spiked with cocaine, there was no basis to disturb hearing officer's conclusion that this only established possibility of unknowing ingestion, but that officer did not meet his burden of persuasion). Because the Department proved that the Respondent ingested and thus possessed cocaine, without any police necessity or authority, the Court finds him Guilty.

PENALTY

In order to determine an appropriate penalty, the Respondent's service record was examined. See Matter of Pell v. Board of Education, 34 N.Y.2d 222, 240 (1974). The Respondent was appointed to the Department on January 19, 1993. Information from his personnel folder that was considered in making this penalty recommendation is contained in the attached confidential memorandum.

The Respondent has been found Guilty of the possession and ingestion of cocaine without police necessity or authority. Accordingly, the Court recommends that he be terminated from employment with the Department. See Disciplinary Case No. 79863/04 (twelve-year member with no prior disciplinary record was terminated from the Department for possessing and ingesting cocaine; two hair samples showed that member deliberately ingested cocaine on multiple occasions); Trotta v. Ward, 77 N.Y.2d 827, 828 (1991) (Police Commissioner's dismissal of officer for use of illegal drugs is not so disproportionate to the offense as to shock one's sense of fairness).

Respectfully submitted,



David S. Weisel
Assistant Deputy Commissioner – Trials

APPROVED



JUN 10 2008
RAYMOND W. KELLY
POLICE COMMISSIONER